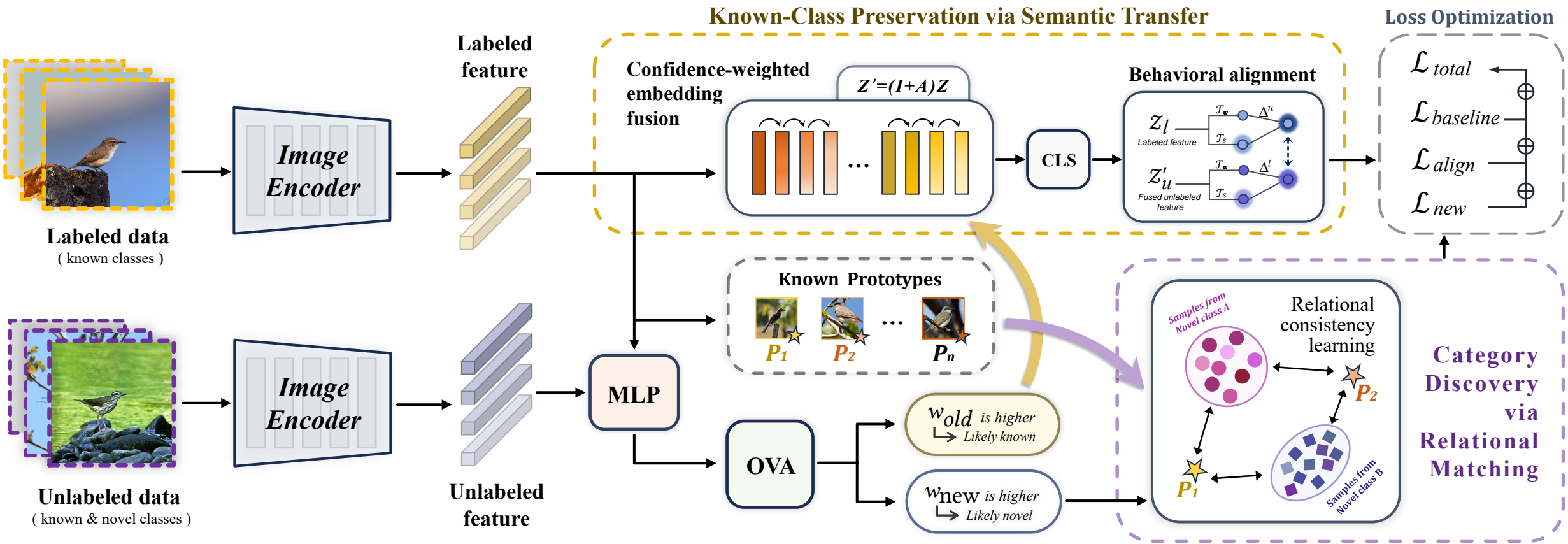
Known-Class Preservation via Semantic Transfer
Loss Optimization
Labeled feature
Confidence-weighted embedding fusion
Z′=(I+A)Z
Behavioral alignment
CLS
Labeled feature
Fused unlabeled feature
Image Encoder
Labeled data
(known classes)
Known Prototypes
P1
P2
Pn
MLP
OVA
w old is higher
Likely known
w new is higher
Likely novel
Image Encoder
Unlabeled data
(known & novel classes)
Unlabeled feature
Samples from Novel class A
Relational consistency learning
Samples from Novel class B
Category Discovery via Relational Matching